Prelim

HSC Practice Questions
Units of Energy and Mass

1 The price and the power consumption of two different brands of television are shown.

Television A	*Television B*
Price: \$900 Power: 176 W	Price: \$921.90 Power: 160 W

The average cost for electricity is 25c/kWh. A particular family watches an average of 3 hours of television per day.

a The annual cost of electricity for Television A for this family is \$48.18.
For this family, what is the difference in the annual cost of electricity between Television A and Television B?
(2 marks) Medium

b For this family, how many years will it take for the total cost of buying and using Television A to be equal to the total cost of buying and using Television B?
(2 marks) Hard

(Q27, **2021 HSC**)

2 Amanda uses 80 kilocalories of energy per kilometre while she is running.

She eats a burger that contains 2180 kilojoules of energy. How many kilometres will she need to run to use up all the energy from the burger? Give your answer correct to one decimal place.
(1 kilocalorie = 4.184 kilojoules) *(2 marks)*

(Q24, **2019 HSC**) Easy

3 The table shows an estimated number of kilojoules burned per kilogram of body mass per 30 minutes in different activities.

Activity	Energy used in 30 minutes
Walking: 6 km/h	9.22 kJ/kg
Cycling: 20 km/h	18.43 kJ/kg
Swimming: 50 m/min	20.73 kJ/kg
Jogging: 10 km/h	21.19 kJ/kg
Running: 16 km/h	32.35 kJ/kg

a Emma has a mass of 58 kg and rides her bike at an average speed of 20 km/h for 90 minutes. What is the amount of energy she has used, to the nearest kilojoule? *(1 mark)* Medium

b James, who weighs 70 kg, drinks a 600-mL bottle of cola which contains 1080 kJ. For how long must James run at 16 km/h to burn off the energy contained in the bottle, to the nearest minute? *(2 marks)* Medium

Bonus question (see page iv)

4 The table shows the number of Calories per 100 grams for three macronutrients.

Source of energy	Number of Calories per 100 grams
Carbohydrate	400
Protein	400
Fat	900

Use the conversion 1 Calorie = 4.184 kJ to find the number of kilojoules in 12.4 g of carbohydrates, to the nearest kilojoule? *(2 marks)*

Bonus question Medium

5 Every day, a 1200-watt microwave oven is used for 45 minutes at 40% power. Electricity is charged at \$0.25 per kWh. What is the cost of running this microwave oven for 180 days? *(3 marks)*

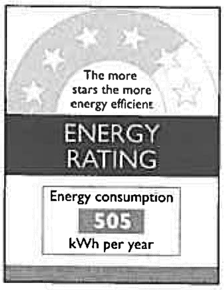
(Q28c, **2018 HSC**) Medium

6 Electricity costs \$0.27 per kWh.

How much does 20 kWh cost? *(1 mark)*

(Q26a, **2017 HSC**) Easy

7 The cost of buying a new heater is \$990. It uses energy according to the following energy label.



Energy is charged at the rate of \$0.35/ kWh.

How much will it cost in total to purchase and then run this heater for five years? *(2 marks)*

(Q28b, **2016 HSC**) Easy

8 The energy consumption of a computer in standby mode is 21 watts. The cost of electricity is 31 cents per kWh.

Units of Energy and Mass

A school computer room has 20 computers.

How much will the school save by switching off all 20 computers during 11 weeks of school holidays? *(2 marks)*

(Q30a, **2015 HSC**) Medium

9 In a household of 4, each member uses an average of 13 minutes of hot water per day.

The household uses a 9 kW hot water unit.

Electricity is charged at 11.97c/kWh when the hot water unit is being used.

What is the electricity cost for the hot water used by this household in one week?

A $1.63 **B** $6.54
C $392.14 **D** $653.56 *(1 mark)*

(Q20, **2014 HSC**) Medium

Units of Energy and Mass

1 **a** Cost for 176 W = \$48.18
Cost for 160 W
= \$48.18 ÷ 176 × 160
= \$43.80 ✓
Difference = \$48.18 − \$43.80
= \$4.38 ✓
(2 marks)

b Difference in price
= \$921.90 − \$900
= \$21.90 ✓
Now, \$21.90 ÷ \$4.38 = 5.
So it will take 5 years for the total cost of both televisions to be the same. ✓
(2 marks)

2 80 kilocalories
= 80 × 4.184 kilojoules
= 334.72 kJ ✓
Distance = (2180 ÷ 334.72) km
= 6.512 9063… km
= 6.5 km (1 d.p.) ✓
(2 marks)

3 **a** Energy used
= 58 × 18.43 × 3
= 3206.82
= 3207 (nearest whole)
∴ Emma will use 3207 kJ of energy. *(1 mark)*

b Total energy per half hour
= 70 × 32.35
= 2264.5 ✓
Number of minutes
= 1080 ÷ 2264.5 × 30
= 14.307 794 22…
= 14 (nearest whole)
∴ it will take 14 minutes. ✓
(2 marks)

4 Number
= 12.4 ÷ 100 × 400 × 4.184 ✓
= 207.5264…
= 208
∴ 208 kJ ✓ *(2 marks)*

5 Power used per day
= 0.4 × 1200 W × $\frac{3}{4}$ h ✓
= 360 Wh
Total power use = 180 × 360 Wh
= 64 800 Wh
= 64.8 kWh ✓
Cost = 64.8 × \$0.25
= \$16.20 ✓ *(3 marks)*

6 Cost = 20 × \$0.27
= \$5.40 *(1 mark)*

7 Energy used in 5 years
= 5 × 505 kWh
= 2525 kWh ✓
Cost of this energy
= 2525 × \$0.35
= \$883.75
Total cost = \$990 + \$883.75
= \$1873.75 ✓
(2 marks)

8 11 weeks = 11 × 7 × 24 hours
= 1848 h
Energy usage
= 20 × 21 W × 1848 h
= 776 160 Wh
= 776.16 kWh ✓
Cost = 776.16 × \$0.31
= \$240.6096
= \$241 (nearest dollar)
So, the school would save around \$241 by switching off the computers. ✓
(2 marks)

9 Total time = 7 × 4 × 13 minutes
= 364 minutes
= 6.066 66… h

Power used
= 9 kW × 6.066 66… h
= 54.6 kWh

Cost = 54.6 × \$0.1197
= \$6.535 62
= \$6.54 (nearest cent)

Answer B